B.Tech Technical Writing Report | Presentation

Does Abliteration Survive 4-bit Quantization?

A Controlled Reproduction and Semantic Evaluation on Qwen3.5-9B



the 32 layers of Qwen3.5-9B; blue layers carry the edit

Omm P Tripathy
B123082

Shreyash Jena
B123124

Aman Saini
B423008

Supervisor: Prof. Srichandan Sobhanayak

Department of Computer Science
International Institute of Information Technology Bhubaneswar

November 2026

MOTIVATION

Edited models get shrunk, but the edit is rarely checked

1

Chat models refuse

They are trained to refuse harmful requests.

2

Abliteration

Deletes one “refusal direction” from the weights, so the model stops refusing.

3

Shrink to 4-bit

Users compress the model so a 9-billion-parameter model fits on one consumer GPU.

?

Still edited?

Whether the edit survives the shrinking is rarely checked.

Earlier work studied how quantization changes a model’s trained safety. We ask a narrower question: does a small, deliberate weight edit survive the same rounding?

RESEARCH QUESTION

One question runs through the whole report

When an abliteration edit is applied at full precision and the model is then shrunk to 4-bit NF4, does the edit's effect survive? And can any loss be told apart from ordinary quantization damage?

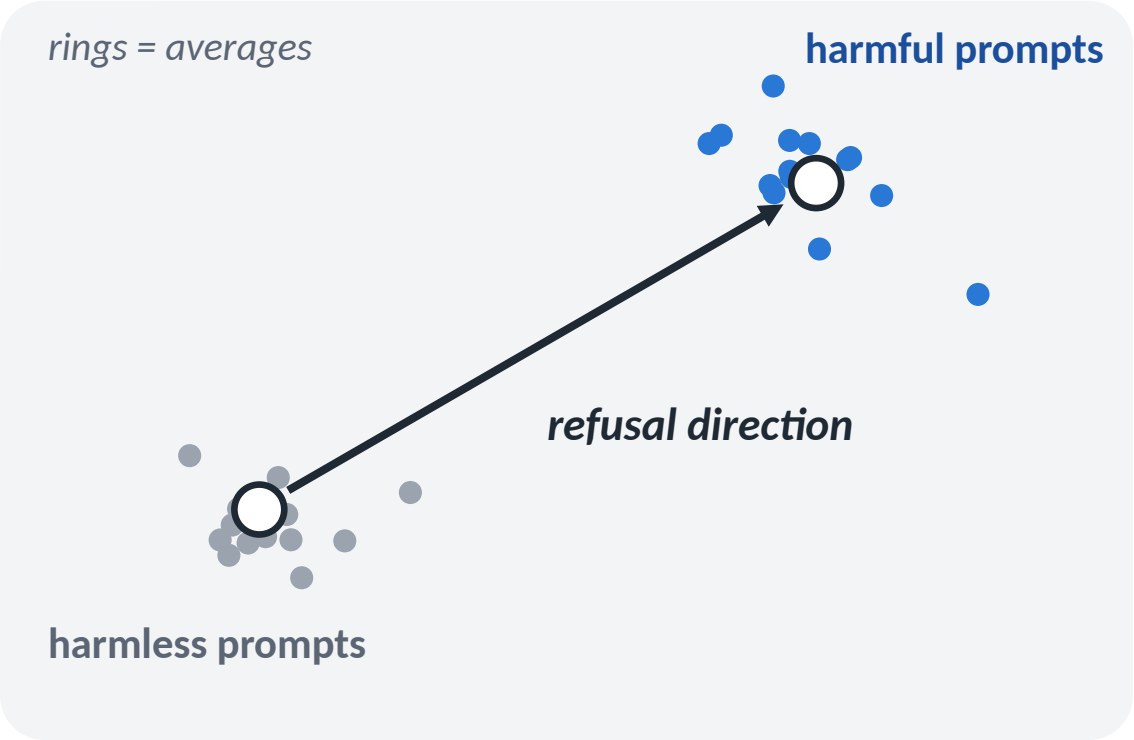
Part 1: survival

After 4-bit conversion, does the edited model still behave like the edited model?

Part 2: attribution

If something changes, was the edit lost, or does 4-bit rounding hurt every model a little?

A refusal direction is the gap between two averages



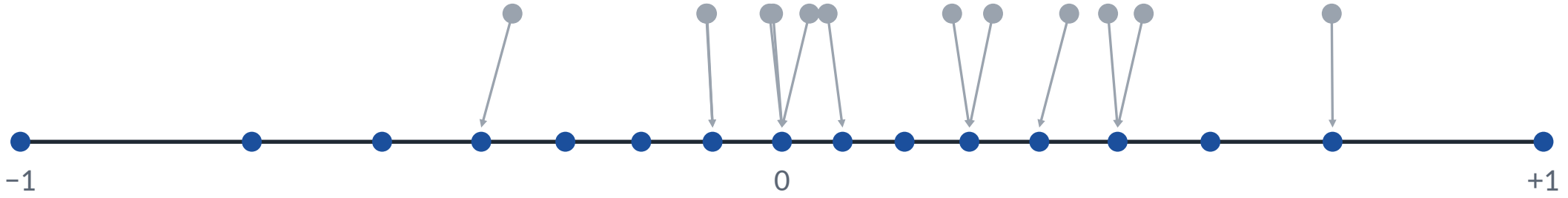
refusal direction = average(harmful) − average(harmless)

1. Run 400 harmful and 400 harmless prompts and record the model's internal numbers (activations) at each layer.
2. Average each group. The arrow from one average to the other is the refusal direction.
3. Edit each layer's output weights so the layer can no longer write along that direction.

BACKGROUND: QUANTIZATION

NF4 rounds every weight to one of 16 levels

original 16-bit weights (any value)



16 NF4 levels, packed tighter near zero where most weights sit

Group

Weights are handled in blocks of 64.

Scale

Each block is divided by its largest absolute value.

Round

Each weight snaps to the nearest of 16 levels: 4 bits.

At run time the weights are expanded back to 16-bit, but the rounding error stays. Could it blur the edit away?

APPROACH

We answer it in three steps; the first two are preconditions

1 Step 1

Get a faithful edit

Can we rebuild the published edit exactly enough to study it?

2 Step 2

Get a trustworthy measure

Does “no refusal phrase” really mean the model answered?

3 Step 3

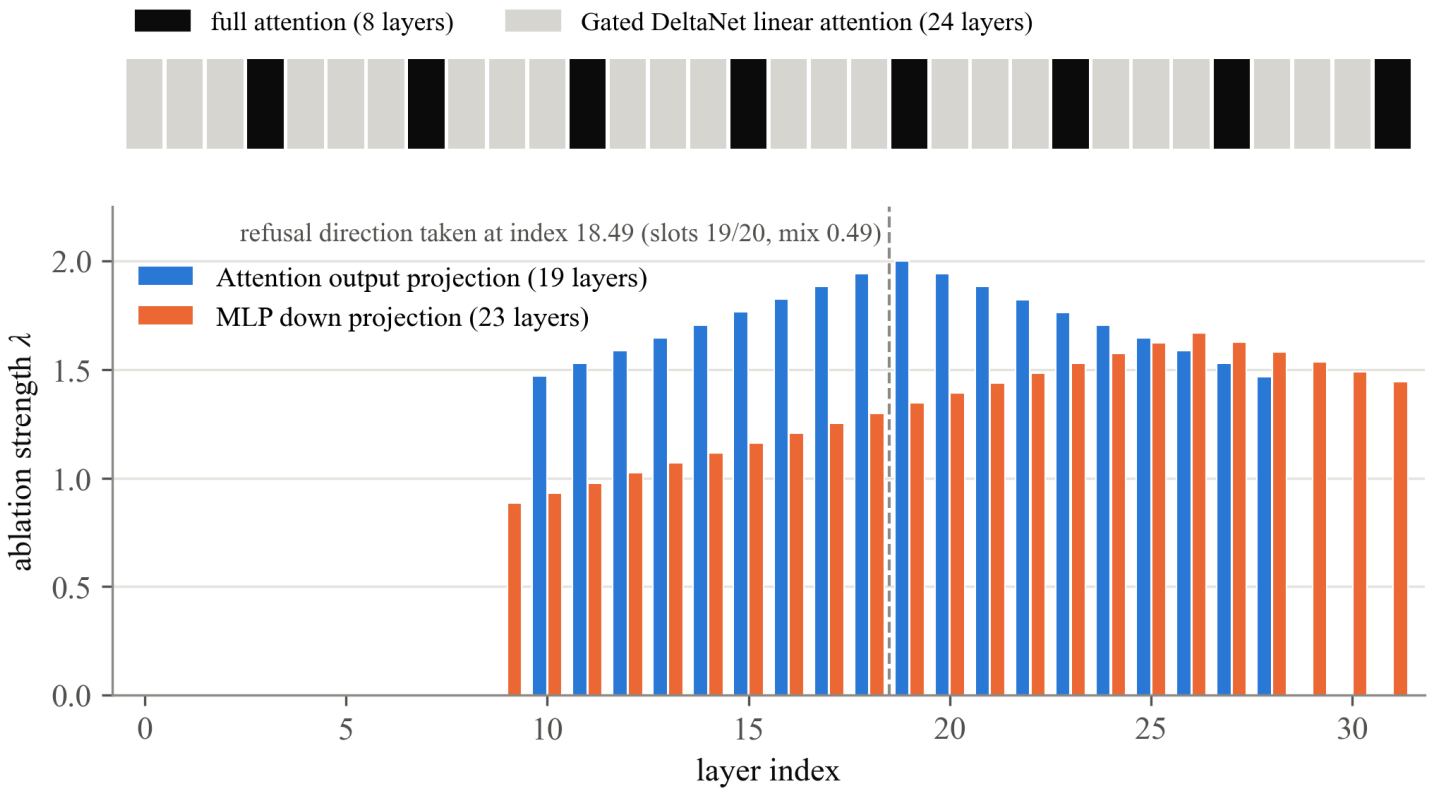
Test survival under NF4

Does the edit’s effect stay after 4-bit conversion, and is any change due to the edit?

Without step 1, a change tells us nothing about the real method. Without step 2, we could not see the change.

SETUP

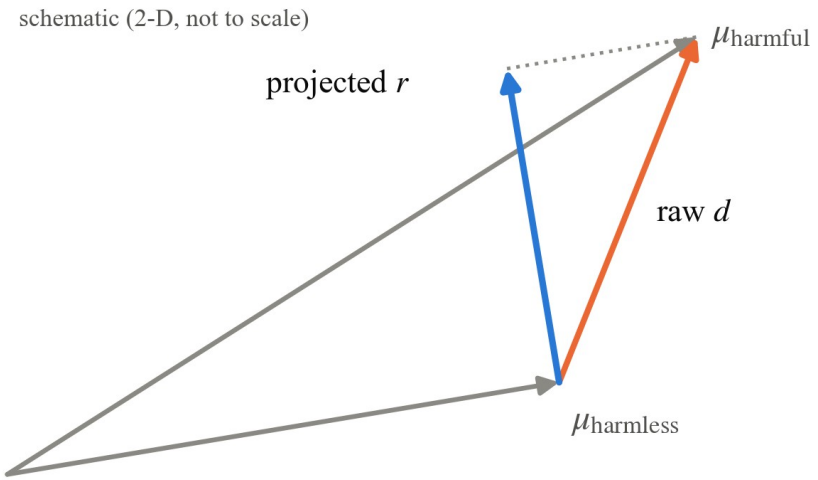
One hybrid model, one published edit, two Kaggle GPUs

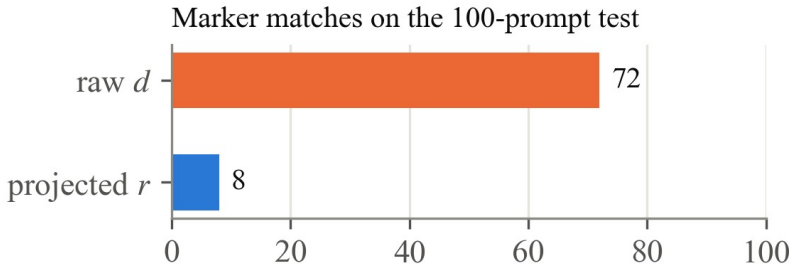


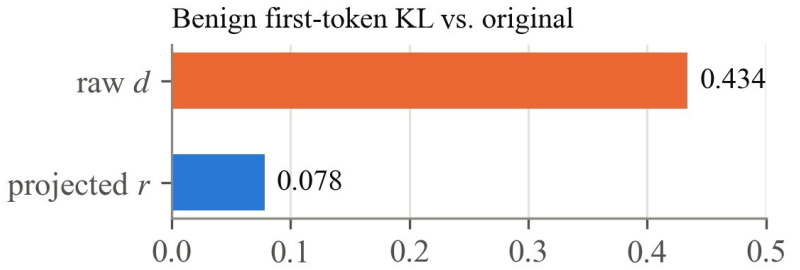
- **Qwen3.5-9B:** 32 layers, three linear-attention layers to every one full-attention layer
- **Published Heretic edit** rebuilt from its released settings, with no new search
- **42 matrices edited:** 19 attention outputs and 23 MLP outputs
- **Compute:** two NVIDIA T4 GPUs on Kaggle, 16-bit

STEP 1 RESULT

The first rebuild failed; one missing step fixed it







Same fixed strengths and layers; only the projection differs.
Controlled check on 32 development prompts: 20/32 → 0/32 markers, KL 0.438 → 0.066.

projected direction = raw direction − its part along the harmless average

Why: every prompt shares the harmless part, so removing it disturbed all inputs and wasted part of the edit.

STEP 1 CHECKS

The rebuilt edit is verified, but phrase counts are not answers

42 of 42

edited matrices equal an explicit computation of original + edit

100 of 100

identical outputs after saving and reloading the model

> 0.9999

cosine match with the publisher's change (MLP and full-attention matrices)

Caution: 8 refusal phrases does not mean 92 answers

With a 100-token limit, 99 of 100 replies were cut off. Whether the model actually answered needs a different measure: that is Step 2.

STEP 2: THE MEASURE

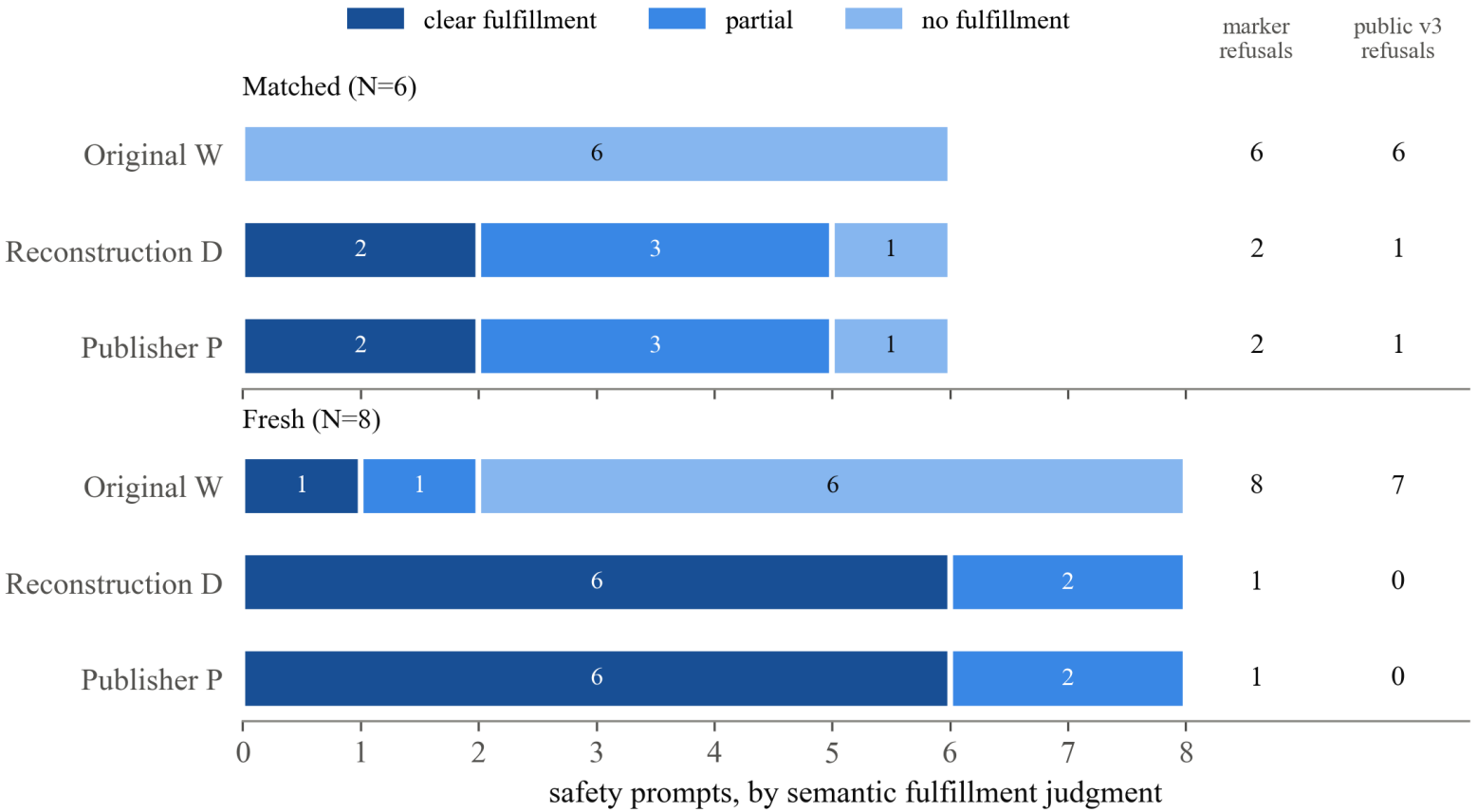
Counting refusal phrases misleads in both directions

	Gave the requested content	Did not give it
Uses refusal words	**Refuses, then answers anyway** Counter says “refusal”: wrong *seen in the original model*	**Plain refusal** Counter is right
No refusal words	**Plain answer** Counter is right	**Polite redirect** Counter says “answered”: wrong *seen in an edited model*

A phrase counter reads the wording. We need to judge what the reply actually gives.

STEP 2 RESULT

Judged by content, the edit clearly works at full precision



6 of 8

fresh safety prompts clearly answered by each edited model

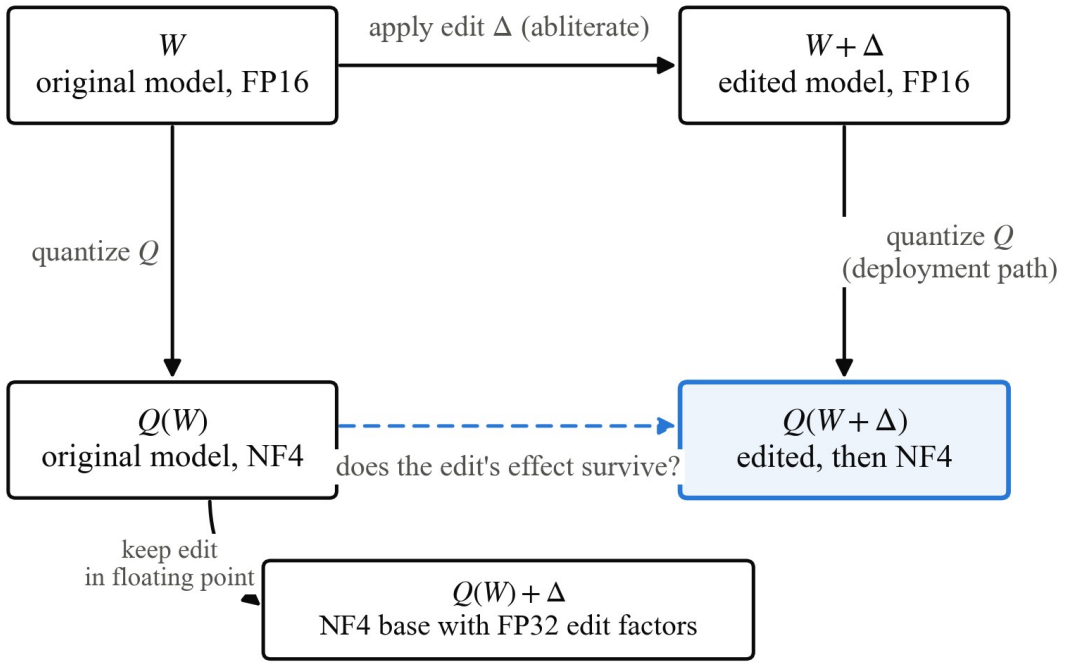
1 of 8

by the original model

Our rebuild and the publisher's model got the same label on all 14 prompts.

STEP 3: THE DESIGN

Five versions separate edit loss from rounding damage



Not run in this study: re-extracting the direction from $Q(W)$.

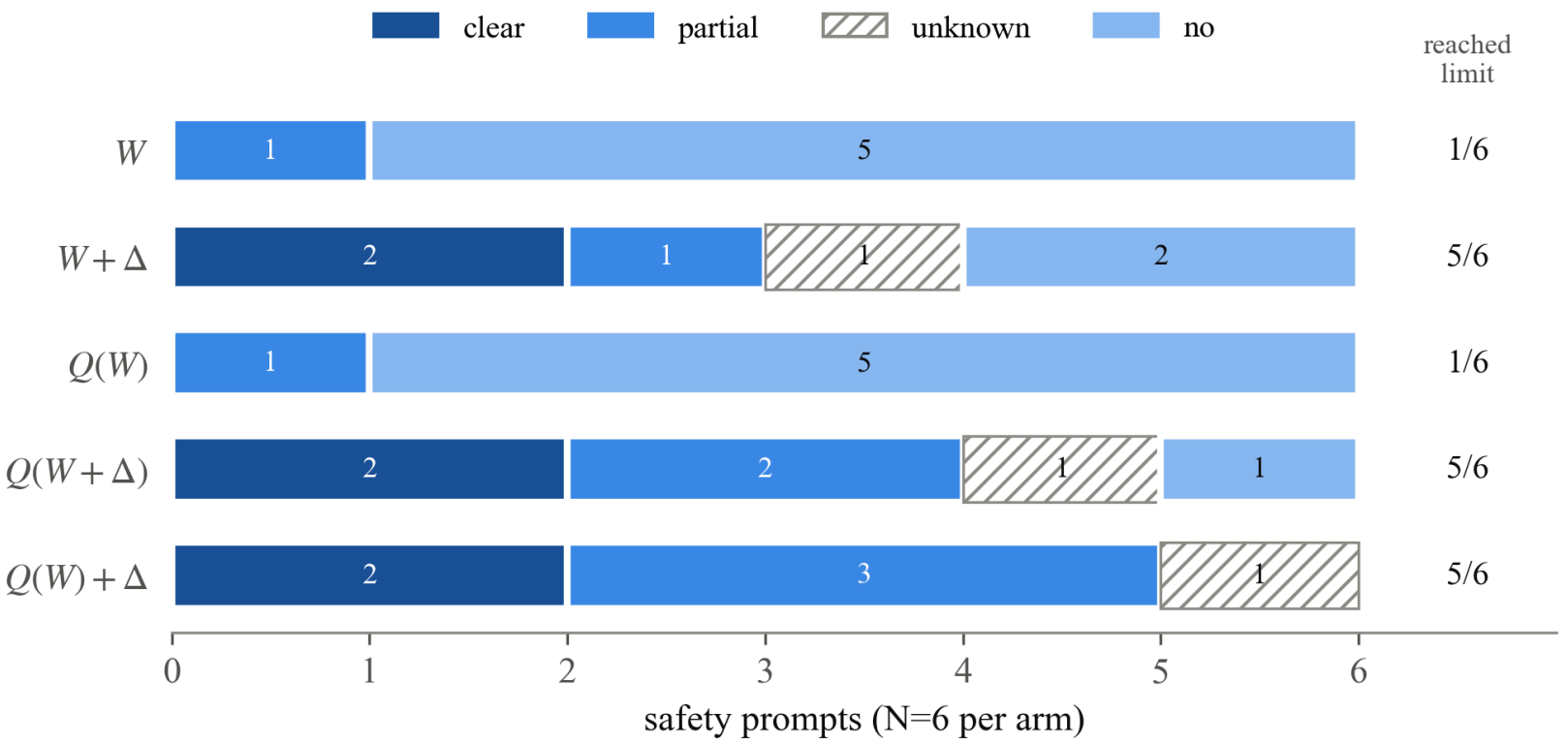
W	original, 16-bit
W + Δ	edited, 16-bit
Q(W)	original, 4-bit: the control
Q(W + Δ)	edited, then 4-bit: deployment
Q(W) + Δ	4-bit base, edit kept unrounded

Why the control matters

If Q(W) started answering too, 4-bit rounding, not the edit, could explain the result.

STEP 3 RESULT

The edit's clear answers survive NF4; NF4 alone creates none



Six safety prompts per condition: a pilot, not a rate.

2 → 2

clear answers before and after NF4

0 → 0

for the original model: NF4 alone creates none

Open

partial weakening not ruled out; most edited replies were cut off

DISCUSSION

Hypothesis, not measured

Why it might survive: a coherent push beats scattered noise

The edit

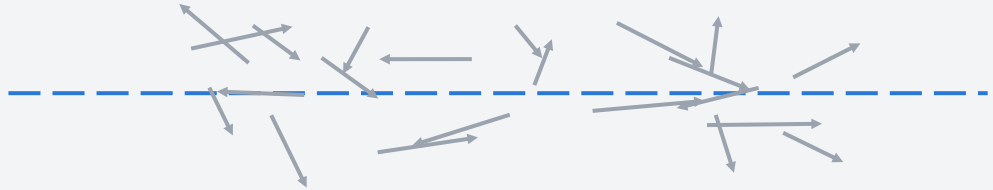
one large push along one direction



refusal direction

Rounding error

many small pushes in every direction



refusal direction

- The edit has strength 0.84 to 2.0, all along one direction. Rounding errors are small and spread over 4096 directions, so only a tiny share lands on the refusal direction.
- Prediction: round-to-nearest 4-bit keeps the edit, unless the strength is small or the format is much coarser.

LIMITATIONS

What this pilot cannot tell us

1 Small samples

14 safety prompts at 16-bit and 6 in the NF4 pilot. No rates or confidence intervals.

2 One AI reviewer

All semantic labels came from one LLM reviewer with imperfect masking. No human labels yet.

3 Truncated answers

Most edited replies hit the token limit, so labels cover only what was visible.

4 One model, one edit

Only Qwen3.5-9B and one published Heretic edit were studied.

5 One quantizer

Only bitsandbytes NF4, in one configuration. Other formats may differ.

6 Not checked

Correctness of harmful content, on purpose. The 100-prompt test was also used while debugging.

We reproduced only public work and show no harmful text

E Ethics

- Only an edit and a checkpoint that were already public were reproduced.
- No harmful model output is shown; replies are described by category only.
- Raw outputs and the rebuilt model are kept private and not released.
- Correctness of harmful answers was deliberately not tested.

AI AI assistance

- AI coding agents helped write the code and run the experiments.
- Semantic labels came from a single LLM-based reviewer.
- All responses and settings were saved for later human review.

CONCLUSION

The edit survived NF4 here; the hard part was measuring it

On Qwen3.5-9B with NF4, the edit's observed effect survived quantization. Whether it is partly weakened remains open at this sample size.

1

Rebuild every step

One missing projection step was the difference between 72 and 8 refusal phrases out of 100.

2

Measure answers, not phrases

Phrase counters err both ways. Judge what each reply actually gives.

3

The edit survived NF4

Both clear answers stayed after 4-bit conversion, and NF4 alone created none.

FUTURE WORK

Three next steps, cheapest first

1

Weight-only retention test

Measure how much of the edit remains in the 4-bit weights, for NF4, 8-bit and lower formats.

runs on a CPU, no text generation

2

Human labels

People label the 132 saved responses, with two reviewers on a shared subset to measure agreement.

checks the AI reviewer

3

Larger study

More prompts, longer answers, a calibrated evaluator such as StrongREJECT, a second quantizer and model.

turns the pilot into rates

Also: re-extract the refusal direction from the 4-bit model, a condition this study did not run.

Thank you

Questions are welcome

72 → 8

refusal phrases out of 100 after the projection fix

6 of 8 vs 1 of 8

fresh prompts clearly answered: edited vs original

2 → 2

clear answers kept after NF4; none created by NF4 alone

Omm P Tripathy
B123082

Shreyash Jena
B123124

Aman Saini
B423008

Supervisor: Prof. Srichandan Sobhanayak
Department of Computer Science, IIIT Bhubaneswar | November 2026

BACKUP

Reproduction numbers on the 100-prompt test

Condition	Refusal-phrase matches	Reached 100-token limit	Benign KL
Original W	100	59	0
Raw-direction replay	72	91	0.434
Publisher P	8	100	0.078
Reconstruction D	8	99	0.078
D after save and reload	8	99	not measured
Publisher’s model card (reported)	6	n/a	0.079

- Controlled check on 32 development prompts: 20 of 32 → 0 of 32 phrase matches; KL 0.438 → 0.066.
- KL: change in the first-token prediction on 100 harmless prompts, compared with the original model.